Neurocomputing 454 (2021) 339–349

Contents lists available at ScienceDirect

Neurocomputing

journal homepage: www.elsevier.com/locate/neucom

Inter-patient ECG arrhythmia heartbeat classification based on unsupervised domain adaptation

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ARTICLE INFO

Article history:
Received 7 June 2020
Revised 6 March 2021
Accepted 26 April 2021
Available online 30 April 2021
Communicated by Zidong Wang

Keywords:
ECG heartbeat classification
Deep learning
Unsupervised domain adaptation

ABSTRACT

Electrocardiography (ECG) arrhythmia heartbeat classification is essential for automatic cardiovascular diagnosis system. However, the enormous differences of ECG signals among individuals and high price of labeled data have brought huge challenges for current classification algorithms based on deep neural networks and prevented these models from achieving satisfactory performance on new data. In order to build a classification system with better adaptability, we propose a novel Domain-Adaptative ECG Arrhythmia Classification (DAEAC) model based on convolutional network and unsupervised domain adaptation (UDA). Based on observation of clustering characteristics of data, we present two original objective functions to enhance the inter-patient performance. A Cluster-Aligning loss is presented to align the distributions of training data and test data. Simultaneously, a Cluster-Maintaining loss is proposed to reinforce the discriminability and structural information of features. The proposed method requires no expert annotations but a short period of unsupervised data in new records to make deep models more adaptive. Extensive experimental results on three public databases demonstrate that our method achieves competitive performance with other state-of-the-arts on the detection of ventricular ectopic beats (V), supraventricular ectopic beats (S) and fusion beats (F). The cross-dataset experimental results also verify the great generalization capability.

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1. Introduction

Electrocardiograph (ECG) plays an essential role in daily cardiovascular healthcare due to its capability of noninvasively recording myocardial electrophysiological activity of the heart [1]. Diagnosing arrhythmia based on ECG requires strong medical expertise. To reduce the cost of diagnosis, researchers have paid great attention to ECG-based automatic arrhythmia diagnosis algorithms in the past few decades. A typical and basic task in automatic ECG arrhythmia diagnosis is heartbeat classification. Correct classification of heartbeats plays an important role in the monitoring of heart health. According to ANSI/AAMI EC57:1998 standard [2], each heartbeat can be classified into one of the five categories (normal beat (N), SVEB (S), VEB (V), fusion beat (F) and unclassifiable beat (Q)). The goal of automatic ECG heartbeat classification is to classify each heartbeat into one of the five categories.

In recent years, deep neural networks have achieved unprecedented development. Due to the powerful capability of feature learning and large-scale annotated clinical data, plentiful deep learning methods have been designed for solving ECG heartbeat classification problem. There are two main types of approaches that are based on convolutional neural networks (CNNs) [3–8] and recurrent neural networks(RNNs) [9–11], respectively. CNNs hierarchically extract multi-scale waveform features, and send the high-level features to classifiers for prediction. RNNs model the time relationships within ECG signals and integrate features at different time steps. Most methods can be classified into either intra-patient paradigm or inter-patient paradigm. Intra-patient methods achieve saturated accuracy on public databases since the training and test data are from the same patients. A more reasonable approach is inter-patient paradigm, where training and test data come from different patients.

Although deep learning methods have made remarkable progress, they still face huge challenges in inter-patient paradigm. The morphological characteristics of ECG signals can vary greatly depending on different individuals, which causes models trained on records from limited individuals severe performance degradation when tested on new records from new patients. The discrepancy between training and test data, referred to as *domain shifts*

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https://doi.org/10.1016/j.neucom.2021.04.104
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[12], may violate the basic independent identically distributed (I.I.D) assumption in learning-based schemes and is common in reality. A natural solution to improve models' generalization ability is to collect and annotate data covering all patients for supervised training. However, annotating ECG records requires strong expertise and is labor-intensive.

Some methods, e.g. patient-specific schemes [13,3,11,14] and active learning [4,15], have been proposed to improve the adaptivity of models to new patients. The patient-specific schemes train a model for each patient, which is space-inefficient. The active learning algorithms learn to pick up informative samples to annotate, making models adaptive to new data with much lower cost. Nevertheless, it is still expensive for heartbeat level classification when the number of new records is large.

To address the issues mentioned above, we propose a novel Domain-Adaptive ECG Arrhythmia Classification (DAEAC) model for improving the inter-patient performance of deep neural networks without additional expert annotation. First, we design a robust model to tackle inter-patient ECG heartbeat classification problem. The new baseline model achieves the state-of-the-art performance on public databases, including MIT-BIH database, SVDB and INCARTDB. Second, based on observation of clustering characteristics of data, we introduce two novel objective functions called *Cluster-Aligning loss* and *Cluster-Maintaining loss*. The Cluster-Aligning loss aligns the distributions of training and test data in the feature space, making the deep models more adaptive to new data. The Cluster-Maintaining loss maintains the structural information and increases the discriminability of new data by using a short period of unlabeled data in test records. The proposed method requires no expert annotation in new records, which is efficient and low-cost.

Our method effectively improves the performance of deep models on new data and does not require any additional human labor, and we maintain only one global model for all patients in the test dataset, which is different from most patient-specific methods that maintain a specific model for each patient and more efficient. The proposed DAEAC is suitable for applications requiring efficient adaptation to new data, such as the online diagnostic system and customized portable device. Experimental results on three public databases demonstrate the effectiveness of our algorithm. Both quantitative and qualitative results prove that the proposed domain adaptation algorithm strikingly reduces the discrepancies of different domains and improves the inter-patient classification performance. The comparison results indicate that our method overperform the state-of-the-arts. The experimental codes will be released on https://github.com/chenmingTHU/DAEAC for further study.

The rest of this paper is organized as follows. Section II introduces existing methods related to our work. Section III gives detailed descriptions of the proposed method. Section IV presents the experimental setups and used databases. Section V shows the experimental results of our algorithm and some quantitative and qualitative analysis. Section VI concludes this work.

2. Related works

2.1. Inter-patient deep learning models

The training and test sets of inter-patient settings contain records from different individuals. In [7], Sellami et al. applied a typical CNN model and proposed a batch-weighted loss function to process three consecutive heartbeats. Zhai et al. [6] transformed heartbeats into a 2-D coupling matrix and used a simple CNN to learn the morphology information as well as the temporal relationship in adjacent beats. Li et al. [8] proposed a 3-D input format and applied a multi-field CNN to achieve high accuracy on public datasets. Zhang et al. [11] viewed ECG heartbeat segmentations as sequential data and applied a two-layer RNN to model the signals. In [5], Guo et al. combined DenseNet [16] and RNN to process long-term ECG segments and achieved impressive performance. Although these methods have achieved impressive results, most of them lack sufficient experiments on multiple datasets, which contain more complex situations. Different from the methods above, our model is specifically designed for enhancing inter-patient performance. Crucially, we use novel objective functions to take advantage of prior information to improve the performance on new datasets.

2.2. Patient-specific models

Patient-specific model [13,3,11,14,17] is a representative strategy to improve the performance of heartbeat classification. The general idea of patient-specific methods is separately establishing a common training set and specific training sets for each patient. A global model is trained on the common training set, and multiple specific models are fine-tuned on specific training sets. A multi-view learning method was proposed in [14] with the usage of pseudo labels and patient-specific models. Ref. [17] proposed I-vector adapted patient-specific DNNs to adjust the patient-independent DNN to specific individuals. Though patient-specific models can achieve high accuracy, they need to save a set of parameters for each patient, which is space-consuming. In addition, some of them generally require considerable labeled data in each new records, which is impractical in real scenario. Our method only maintains a global model for all patients, and we requires no labels on new records for training.

2.3. Unsupervised domain adaptation

Unsupervised domain adaptation (UDA) [18] is an important research issue in machine learning. In UDA, the distributions of training data and test data are generally considered to be related, but not identical. The training data is referred to as *source domain*, and the test data is referred to as *target domain*. The idea is to use unlabeled data in the target domain to minimize the distribution discrepancy between two domains in sample space or feature space. In other words, UDA algorithms guide the deep networks to learn *domain-invariant* feature representations.

The advantage of UDA is that it does not require additional annotations and can be easily embedded into deep learning models. Some UDA methods have been applied in ECG monitoring [19–21] and ECG classification [22,23]. Ref. [22] used domain transfer SVM and importance weighted kernel logistic regression method to enhance the classification performance by minimizing the Maximum Mean Distribution (MMD) between two distributions. Ref. [23] proposed a semi-supervised system based on a 2D convolutional neural network with inputs generated from heartbeats triplets. However, experiments across datasets were not conduct in precursory works. In our work, we apply UDA to tackle ECG heartbeat classification problem in an end-to-end manner without any structure modification to original neural networks. We conduct extensive experiments on multiple datasets to vertify the effectiveness.

3. Domain-adaptive ECG arrhythmia classification

3.1. Framework

A diagram illustrating the concept of domain adaptation is shown in Fig. 1. Generally, we refer the training dataset as *source*

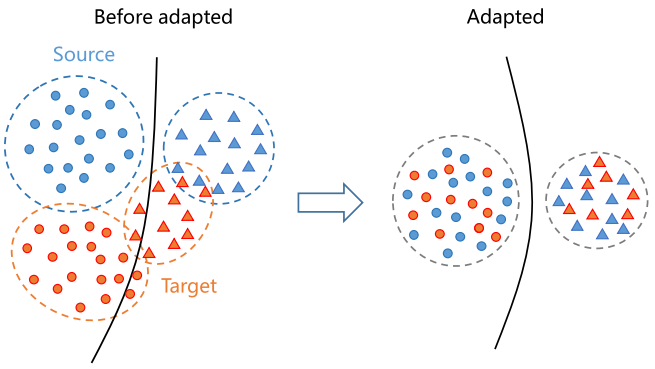


Fig. 1. Left: The source and target domains have discrepancy, which is referred as *domain shifts* and may cause performance degradation in targte domain using the classifier trained on source domain. Right: Our algorithm can align the distributions of two domains after adaptation.

domain and the test dataset as *target domain*. The source domain has samples $\mathcal{X}_s = \{x_s^i\}_{i=1}^N$ with labels $\mathcal{Y}_s = \{y_s^i\}_{i=1}^N$, while the target domain has samples $\mathcal{X}_t = \{x_t^i\}_{i=1}^M$ without labels. Our goal is to train the models with $\{\mathcal{X}_s, \mathcal{Y}_s, \mathcal{X}_t\}$ to performs as well as possible on the entire test dataset. For the heartbeat classification problem, a model consists of a feature extractor $\mathcal{F}$ and a classifier $\mathcal{H}$. $\mathcal{F}$ maps a heartbeat sample into a feature vector according to which $\mathcal{H}$ predicts the label.

The framework of the proposed algorithm can be viewed in Fig. 2. The proposed algorithm is composed of two stages: the pre-training stage and adaptation stage. In the pre-training stage, the feature extractor $\mathcal{F}$ and classifier $\mathcal{H}$ are optimized with the labeled source data $\{\mathcal{X}_s, \mathcal{Y}_s\}$ in a supervised manner to obtain a decent heartbeat classifier. In the adaptation stage, in order to bridge the gap between source and target domain, both source data $\{\mathcal{X}_s, \mathcal{Y}_s\}$ and target data $\mathcal{X}_t$ are used to optimize $\mathcal{F}$ by minimizing the discrepancy between two domains in the feature space.

3.2. ECG heartbeat classification model

To design a better $\mathcal{F}$ and $\mathcal{H}$, we first propose a robust baseline CNN model to reach comparable performance with other state-of-the-arts. The detailed structure of the model is shown in Fig. 3. The design is inspired by [8], while we modify the structure and input to make the model more suitable for our objective. As can be seen in Section IV, although most of the existing deep models have achieved high accuracy, these models have variant performance on different classes, which is a serious problem for domain adaptation. Our model can achieve competitive overall accuracy with more balanced performance.

3.2.1. Network inputs

Similar to [8], we apply a special input structure that supports both morphological and temporal information. We simplify the input format by using single-lead signals instead of two-lead signals and removing beat-to-beat correlation, since we want the data to be as accessible and versatile as possible, e.g., data collected by wearable devices.

A schematic of the structure of the inputs is shown in Fig. 4. The input tensor consists of three parts: a single heartbeat ECG segment, pre-RR ratios and near-pre-RR ratios. Each segment starts at the 0.14s after the previous R-peak, and ends at the 0.28s after current R-peak. The pre-RR ratio is the ratio of the current pre-RR interval to the average of all pre-RR intervals before the current heartbeat. The near-pre-RR ratio is the ratio of the current pre-RR interval to the average of last ten pre-RR intervals. Since both the pre-RR ratio and near-pre-RR ratio are scalars, we repeat them to the same length as the ECG segment and then concatenate them together to form the inputs.

The sampling frequency of ECG in different data sets is different (360 Hz, 257 Hz and 128 Hz for MIT-BIH, INCARTDB and SVDB, respectively). To solve the issue of diverse sampling rates, in this work, we use FFT-based resampling algorithm to first unify the sampling rates to 360 Hz before training and evaluation. After segmentation, all ECG segments are resampled to a fixed length 128.

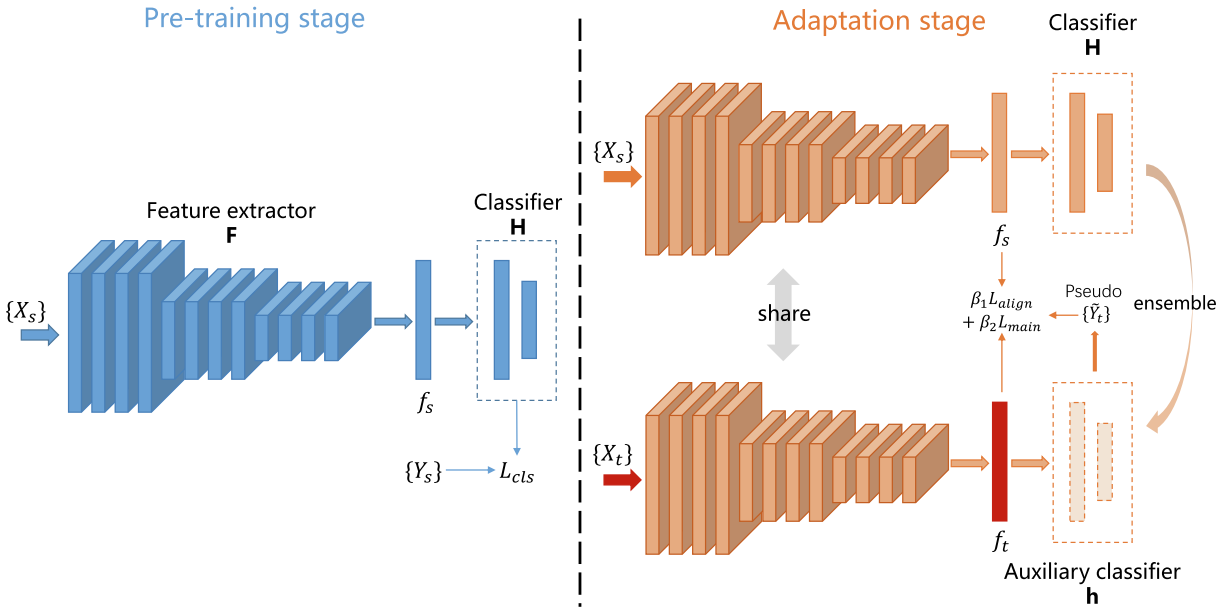


Fig. 2. (Best viewed in color) The framework of proposed DAEAC model. The algorithm consists of two stages: pre-training stage and adaptation stage. In the pre-training stage, the feature extractor $\mathcal{F}$ and classifier $\mathcal{H}$ are optimized according to standard cross entropy loss with labeled source data. In the adaptation stage, unlabeled target data are involved to update the feature extract $\mathcal{F}$, guiding it to learn domain-invariant features. Simultaneously, the classifier $\mathcal{H}$ is also updated with labeled source data. The auxiliary classifier h has the same struture with $\mathcal{H}$, which is applied to generate pseudo labels in the adaptation stage. The h is updated with $\mathcal{H}$ at the beginning of each epoch.

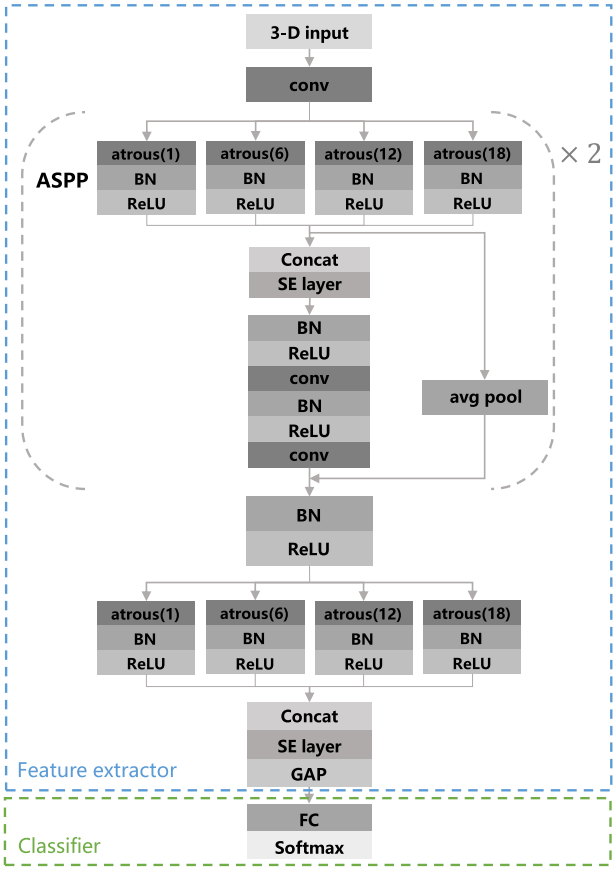


Fig. 3. The structure of proposed network in DAEAC. **ASPP**, **atrous** and **SE** refer to atrous spatial pyramid pooling module, atrous convolution and Squeeze-and-Excitation layer, respectively.

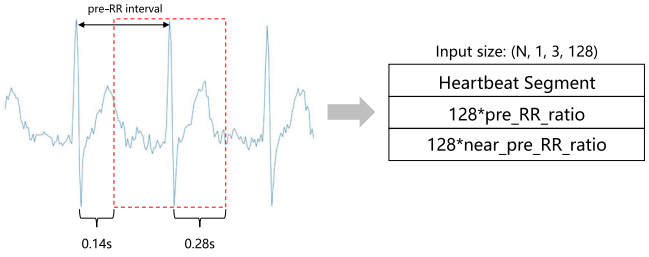


Fig. 4. The input structure of our network. An input tensor is composed of a single heartbeat ECG segment, pre-RR ratios and near-pre-RR ratios. The ECG segment provides morphological information, and pre-RR and near-pre-RR ratios provide temporal information.

3.2.2. Network architecture

Our model is proposed based on CNN. The entire network can be viewed as two components: a feature extractor $\mathcal{F}$ and a classifier $\mathcal{H}$. Multiple atrous spatial pyramid pooling (ASPP) modules[24] at different level to extract multi-field features. Additionally, we add a Squeeze-and-Excitation (SE) layer [25] after each ASPP module. The intuitive idea is that the feature channels are not entirely equivalent, and SE layers can automatically re-weigh the feature channels extracted by the ASPP module during the training process. In addition, this feature re-weighting mechanism is naturally suitable for the domain adaptation problem since the SE layers may be able to suppress the features that are not related to the task but specific to the dataset. We use three ASPP-SE modules at different levels in the network to reduce the redundancy of the learned features. After multiple convolutional layers and a global average pooling layer (GAP), $\mathcal{F}$ maps an input tensor into a 256-D feature vector. The classifier $\mathcal{H}$ consists of a fully connected layer and a softmax layer, which produces a probability for each category (N, S, V, F in this work, and Q is not extended).

3.3. Unsupervised domain adaptation

Algorithm 1. Unsupervised Domain Adaptation.

Require
Source samples and labels: $\mathcal{X}_s$, $\mathcal{Y}_s$;
Target samples: $\mathcal{X}_t$;
Feature extractor $\mathcal{F}$, classifier $\mathcal{H}$ and auxiliary classifier h;
Epochs, β_1, β_2, source size N_s, target size N_t, batch size N; γ
Ensure
Trained model: $\mathcal{F}$, $\mathcal{H}$;
for $t = 1$ to *Epochs* **do**
  Compute weighted cross entropy: $\mathcal{L}_{cls} = -\frac{1}{N}\sum_{i=1}^{N} w_i y_i log(\mathcal{H}(\mathcal{F}(x_i^s)))$;
  Update $\mathcal{F}$ and $\mathcal{H}$ with $\mathcal{L}_{cls}$;
end for
Updating h with $\mathcal{H}$;
Obtaining pseudo labels: $\bar{\mathcal{X}}_t$, $\bar{\mathcal{Y}}_t$;
  for $k = 1$ to K
  // Compute initial global clustering centers
  $C_k^s(0) = \frac{1}{N_s}\sum_{i=1}^{N_s}\mathcal{F}\left(x_{k,i}^s\right)$;
  $C_k^s(0) = \frac{1}{N_t}\sum_{i=1}^{\bar{N}_t}\mathcal{F}\left(\bar{x}_{k,i}^t\right)$;
end for
for *epoch* $= 1$ to *Epochs* **do**
  for $t = 1$ to $\frac{N_t}{N}$ **do**
    // Update clustering centers
    for $k = 1$ to K
      $\overline{C}_k^s(t) = \frac{1}{n_k^s}\sum_{i=1}^{n_k^s}\mathcal{F}\left(x_{k,i}^s\right)$
      $\overline{C}_k^s(t) = \frac{1}{\bar{n}_k^t}\sum_{i=1}^{\bar{n}_k^t}\mathcal{F}\left(\bar{x}_{k,i}^t\right)$
      $C_k^s(t) \leftarrow (1-\gamma)C_k^s(t-1) + \gamma\overline{C}_k^s(t)$
      $C_k^t(t) \leftarrow (1-\gamma)C_k^t(t-1) + \gamma\overline{C}_k^t(t)$
      $C_k(t) = \frac{C_k^s(t)+C_k^t(t)}{2}$
    end for
    Compute weighted cross entropy:
    $\mathcal{L}_{cls} = -\frac{1}{N}\sum_{i=1}^{N} w_i y_i log(\mathcal{H}(\mathcal{F}(x_i^s)))$;
    Compute Cluster-aligning loss: $\mathcal{L}_{align} = \sum_{k=1}^{K}\mathcal{D}(C_k^s(t), C_k^t(t))$;
    Compute separating loss:
    $\mathcal{L}_{sep} = \sum_{k\neq l}^{K}\sum_{l=1}^{K} max(\mathcal{T}_m - \mathcal{D}(C_k(t), C_l(t)), 0)$;
    Compute compacting loss: $\mathcal{L}_{comp} = \sum_{k=1}^{K}\sum_{j=1}^{n_k^s+\bar{n}_k^t}\mathcal{D}(x_{k,j}, C_k(t))$
    Compute total loss: $\mathcal{L} = \mathcal{L}_{cls} + \beta_1\mathcal{L}_{align} + \beta_2(\mathcal{L}_{sep} + \mathcal{L}_{comp})$
    Update $\mathcal{F}$ with $\mathcal{L}$; update $\mathcal{H}$ with $\mathcal{L}_{cls}$
  end for
  Updating h with $\mathcal{H}$;
  Update pseudo labels: $\bar{\mathcal{X}}_t$, $\bar{\mathcal{Y}}_t$;
end for

Convolutional neural networks have an implicit property. As the number of layers deepens, the features extracted by the network will become more specific to task and dataset [26]. Most machine learning and deep learning algorithms in ECG classification assume that training data and test data are independently and identically distributed. In practice, ECG signals may have considerable variation among different individuals, which may violate the assumption of independent and identical distribution. To address this problem, we propose two simple but effective loss functions to regularize the network to learn domain-invariant features.

3.3.1. Cluster-aligning loss

A fundamental assumption in the classification task is cluster hypothesis [27]. Samples from the same category should be in the same cluster, while samples belonging to different categories should be far away from each other. Following this underlying

assumption, we propose a Cluster-aligning loss (Eq. 1) to minimize the distance between clusters of the same category in the source and target domain.

$$\mathcal{L}_{align} = \sum_{i=1}^{K} \mathcal{D}\left(\mathbb{E}\left[\mathcal{F}\left(X_i^s\right)\right], \mathbb{E}\left[\mathcal{F}\left(X_i^t\right)\right]\right) \tag{1}$$

As is shown in Eq. 1, the Cluster-aligning loss $\mathcal{L}_{align}$ is used to minimize the distance between centres of corresponding clusters. Where $\mathcal{D}(,)$ is a distance metric function, and K is the number of categories. We use the expectation of each category of samples to represent the entire cluster, and reduce the domain shifts by aligning the cluster centres of all categories in two domains. It should be noted that there are no labels available for the data in the target domain. To obtain the cluster of each category in the target domain, we employ the pre-trained model from the first stage to generate pseudo labels for the unlabeled data. An auxiliary classifier h is applied in our work. The h has the same structure with $\mathcal{H}$ and is updated with $\mathcal{H}$ at the beginning of each epoch in the adaptation stage. In order to reduce the negative impact of incorrect pseudo labels, we select samples with high prediction confidence to calculate the loss functions. Specifically, we only select samples with prediction probability higher than threshold t_i for training. Generally, we can use the same threshold for all categories, but the datasets of ECG heartbeat classification task have serious data imbalance issue. To better adapt to the problem, we set different thresholds for each type. We use 0.99 for S, V, F and 0.999 for N, since the number of type N is significantly larger than the others. The thresholds are selected experimentally on MIT-BIH dataset and applied for all experiments.

3.3.2. Cluster-maintaining loss

While reducing the domain shifts, we hope the strutural information and discriminativity of target domain can be well preserved. To achieve this purpose, we propose an additional regularizer called Cluster-Maintaining loss (Eq. 2) to simultaneously increase the inter-class spacing and reduce the intra-class spacing of samples.

$$\begin{aligned}\mathcal{L}_{main} &= \mathcal{L}_{sep} + \mathcal{L}_{comp} \\ &= \sum_{j\neq i}^{K}\sum_{i=1}^{K} max\left(0, \mathcal{T}_m - \mathcal{D}\left(\mathbb{E}[\mathcal{F}(X_i)], \mathbb{E}\left[\mathcal{F}\left(X_j\right)\right]\right)\right) \\ &\quad + \sum_{k=1}^{K}\sum_{j=1}^{n_k} \mathcal{D}\left(X_{kj}, \mathbb{E}[X_k]\right),\end{aligned} \tag{2}$$

where n_k is the number of samples in *kth* category. The objective function has two items. The first item, called *seperating loss*, is responsible for maximizing the inter-class spacing. And the second item, called *compacting loss*, is responsible for minimizing the intra-class spacing. Different from $\mathcal{L}_{align}$, we compute $\mathcal{L}_{sep}$ and $\mathcal{L}_{comp}$ using the clustering centres of the mixture of source data and target data since $\mathcal{L}_{align}$ gradually align the two domains during the training process. As shown in the formula, $\mathcal{L}_{sep}$ keeps the clustering centres of different classes away from each other to reach a pre-defined threshold $\mathcal{T}_m$, while $\mathcal{L}_{comp}$ keeps all samples in the same cluster as close to the centre as possible.

3.4. Optimization

In this section, we introduce the implementation details of the proposed algorithm. The algorithm consists of two stages, pre-training stage and adaptation stage. The pre-training stage is a standard classification task training process. The source data is used to optimize the feature extractor $\mathcal{F}$ and softmax classifier $\mathcal{H}$. We use a weighted cross-entropy loss (Eq. 3) and and Adam optimizer [28] in the training procedure.

$$\mathcal{L}_{cls} = -\frac{1}{N}\sum_{i=1}^{N} w_i y_i \log\left(\mathcal{H}\left(\mathcal{F}\left(x_i^s\right)\right)\right) \tag{3}$$

In the adaptation stage, we first use the pre-trained $\mathcal{F}$ and $\mathcal{H}$ to obtain pseudo labels for target domain. Since we use thresholds to select samples with high prediction confidence, we can obtain a set of "labeled" target samples $\tilde{\mathcal{X}}_t = \{x_i^t\}_{i=1}^{\tilde{N}_t}, \tilde{\mathcal{Y}}_t = \{\tilde{y}_i^t\}_{i=1}^{\tilde{N}_t}$. Then we can compute the initial clustering centers of all categories for both domains. Ideally, global clustering centers are required for calculating $\mathcal{L}_{align}$ and $\mathcal{L}_{main}$, which is computationally inefficient. In our experiments, we use exponential moving average to gradually update the global clustering centers with the local centers of each mini-batch. The ultimate objective function is the weighted sum of $\mathcal{L}_{cls}$ and the regularizers:

$$\mathcal{L} = \mathcal{L}_{cls} + \beta_1 \mathcal{L}_{align} + \beta_2 \mathcal{L}_{main}, \tag{4}$$

where β_1 and β_2 are combination coefficients, which are both set to small values to balance the magnitude of cross-entropy loss and the proposed regular terms. In our experiments, we empirically set the value to 0.1, and the performance is not sensitive to slight changes around 0.1 of both β_1 and β_2. After each epoch in the adaptation stage, the pseudo labels are updated together with the model during the training process. The complete algorithm is shown in Algorithm 1.

4. Experimental setups

4.1. Databases

We evaluate the performance of proposed algorithm on three different public databases, which have been widely used for evaluation in arrhythmia heartbeat classification task, for comparison with other state-of-the-art methods. The numbers of samples in three databases are shown in Table. 1. Following previous works [4,15], we ignore type Q since there are extremely few positive samples compared to other types in all three databases. E.g. there are only 8 samples in MIT-BIH DS1 and 7 samples in DS2.

4.1.1. MIT-BIH arrhythmia database (MITDB)

MIT-BIH Arrhythmia Database contains 48 records from 47 patients. Each record has two-lead signals of 30 min that are band-pass filtered at 0.1–100 Hz and sampled at 360 Hz. According to ANSI/AAMI convention [2], 4 paced records(102, 104, 107, 217) are excluded from training and test sets. The rest of 44 records are split into two subsets without intersection. DS1 (101, 106, 108, 109, 112, 114, 115, 116, 118, 119, 122, 124, 201, 203, 205, 207, 208, 209, 215, 220, 223, 230) is generally used for training and DS2 (100, 103, 105, 111, 113, 117, 121, 123, 200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232, 233, 234) is used for evaluation.

4.1.2. St. Petersburg Institute of Cardiological Technics 12-lead Arrhythmia Database (INCARTDB)

St. Petersburg Institute of Cardiological Technics 12-lead Arrhythmia Database consists of 75 records with 12 standard leads. Each record is 30 min long and sampled at 257 Hz.

4.1.3. MIT-BIH Supraventricular Arrhythmia Database (SVDB)

MIT-BIH Supraventricular Arrhythmia Database contains 78 half-hour ECG records chosen to supplement the examples of supraventricular arrhythmias in the MIT-BIH Arrhythmia Database. Each record is sampled at 128 Hz.

Table 1
The number of samples in the public databases.

Database	Total Number	N	S	V	F	Number of Records
MITDB	100420	89605	3015	6998	802	44
INCARTDB	175575	153333	2068	19972	202	75
SVDB	184023	161901	12177	9922	23	78

4.2. Experimental procedure

A concise experimental flowchart is shown in Fig. 5. We first obtain a pre-trained model trained only on labeled dataset of source domain, called DAEAC-base, to serve as the baseline. Then we adapt the model on both labeled source data and unlabeled target data with the proposed algorithm to obtain the adapted model called DAEAC-uda. Evaluation results of both DAEAC-base and DAEAC-uda are used for comparison and analysis.

Our experiments are divided into two parts. First, we evaluate the effectiveness of our algorithm on MIT-BIH dataset. Then, cross-dataset experiments are conducted for further verification of the generalization capability. In the first part, we follow the procedure of most research, training the model on DS1 of MITDB and evaluating on DS2. In this experiment, we consider DS1 as the source domain and DS2 as the target domain. For domain adaptation, following the previous work [3], we select the first five minutes of ECG signals of all records in DS2 as the unlabeled set for fine-tuning the pre-trained model. In the second part, we respectively consider DS1 and the entire MITDB (M) as the source domain, and treat other datasets (S short for SVDB, and I short for INCARTDB) as the target domain.

4.3. Evaluation metrics

Four standard metrics are used to evaluate the classification performance: *sensitivity (Se)*, *positive predictivity (Pp)*, *F1 score* and *overall accuracy (Acc)*. The four metrics are defined as follow:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \tag{5}$$

$$Se = \frac{TP}{TP + FN} \tag{6}$$

$$Pp = \frac{TP}{TP + FP} \tag{7}$$

$$F1 = \frac{2 \times Se \times Pp}{Se + Pp} \tag{8}$$

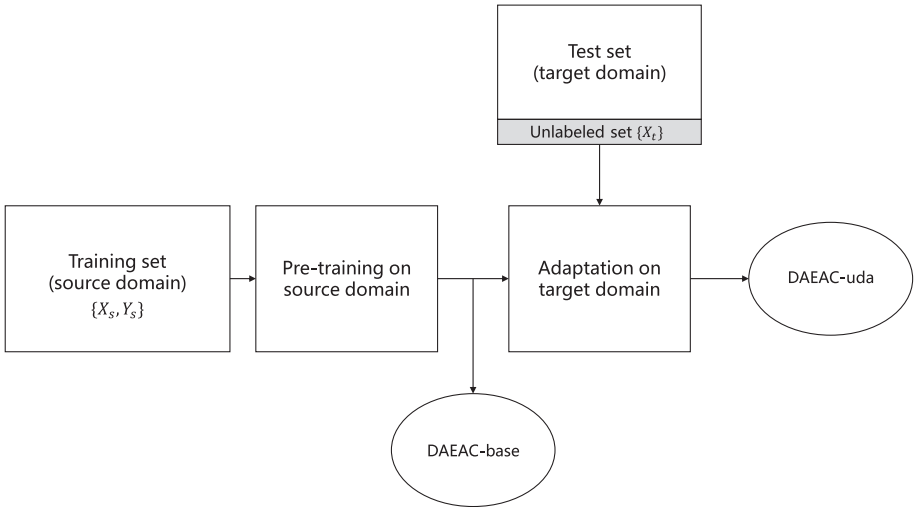


Fig. 5. The experimental flowchart in this work. DAEAC-base is a pre-trained model trained only on labeled dataset of source domain, which serves as the baseline. DAEAC-uda is the adapted model with the proposed unsupervised domain adaptation algorithm. Evaluation results of both DAEAC-base and DAEAC-uda are used for comparison and analysis.

Table 2
Comparisons of the proposed method with other methods on MIT-BIH DS2.

Methods	Types	N			VEB			SVEB			F			Overall accuracy
		Se	Pp	F1	Se	Pp	F1	Se	Pp	F1	Se	Pp	F1	
Huang et al. [29]	Inter-patient	99.2%	95.2%	0.9	93.9%	90.9%	0.92	91.1%	42.2%	0.58	-	-	-	93.8%
Guo et al. [5]	Inter-patient	-	-	-	91.25%	88.30%	0.90	62.70%	61.21%	0.62	–	–	–	–
Niu et al. [30]	Inter-patient	98.9%	97.4%	0.98	85.7%	94.1%	0.89	76.5%	76.6%	0.76	0.25%	1.79%	0.00	96.4%
Sellami et al. [7]	Inter-patient	88.51%	98.80%	0.93	92.05%	72.13%	0.80	82.04%	30.44%	0.44	68.30%	26.58%	0.38	95.33%
Li et al. [8]	Inter-patient	91.81%	98.92%	0.95	95.15%	90.11%	0.92	89.05%	35.41%	0.50	32.22%	20.36%	0.25	91.44%
Zhai et al.[6]	Patient-specific	97.64%	98.51%	0.98	93.8%	92.4%	0.93	76.8%	74.0%	0.75	79.58%	62.36%	0.70	96.14%
Ye et al. [14]	Patient-specific	99.7%	97.2%	0.98	91.8%	98.0%	0.95	61.4%	90.7%	0.73	9.0%	28.3%	0.14	97.0%
Bazi et al. [22]	UDA	–	–	–	–	–	–	–	–	–	–	–	–	93%
DAEAC-base(ours)	Inter-patient	96.06%	98.13%	0.97	92.01%	80.16%	0.86	71.92%	82.89%	0.77	62.64%	23.32%	0.34	94.62%
DAEDA-uda(ours)	UDA	99.11%	98.42%	0.99	94.02%	92.29%	0.93	76.52%	90.19%	0.83	57.86%	74.10%	0.65	**97.59%**

TP, FP, TN and FN represent true positive, false positive, true negative and false negative, respectively. We compare our method with several latest deep learning models designed for ECG heartbeat classification problem [29,5,30,7,8] and adaptative methods [6,14,22]. As for cross-dataset comparison, we compare our model with several methods that had the same setting with us [31,15,4,5].

4.4. Implementation details

We use PyTorch framework to implement our algorithms. In all experiments, we use Adam optimizer to update the parameters of the model. We set the initial learning rate and weight decay to 0.005 and 0.0001, respectively. We decay the learning rate by mul-

Table 3
Comparisons of the proposed method with other methods on INCARTDB.

Method	Labeled Training Dataset	VEB			SVEB		
		Se	Pp	F1	Se	Pp	F1
Rahhal et al. [4]	MITDB	75.1%	37.6%	0.50	15.6%	2.5%	0.01
Wang et al. [15]	MITDB	83.9%	93.0%	0.88	78.3%	22.6%	0.35
DAEAC-base(ours)	MITDB DS1	77.02%	92.08%	0.84	44.07%	37.38%	0.40
DAEAC-uda(ours)	MITDB DS1	90.1%	90.3%	**0.90**	71.1%	43.5%	0.54
DAEAC-base(ours)	MITDB	82.79%	96.87%	0.89	71.65%	32.05%	0.44
DAEAC-uda(ours)	MITDB	82.53%	95.26%	0.88	66.72%	47.60%	**0.55**

* Expert assisted.

Table 4
Comparisons of the proposed method with other methods on SVDB.

Method	Labeled Training Dataset	VEB			SVEB		
		Se	Pp	F1	Se	Pp	F1
Rahhal et al. [4]	MITDB	65.2%	9.31%	0.16	8.8%	14.3%	0.11
Wang et al. [15]	MITDB	85.7%	48.3%	0.62	25.4%	36.7%	0.30
Wang et al. [15]	MITDB + INCARTDB	88.3%	56.3%	0.69	18.4%	36.4%	0.24
Guo et al. [5]	MITDB	86.81%	58.82%	**0.70**	7.90%	64.50%	0.14
DAEAC-base(ours)	MITDB DS1	84.02%	37.41%	0.52	14.16%	39.20%	0.21
DAEAC-uda(ours)	MITDB DS1	85.72%	51.64%	0.64	20.47%	53.82%	0.30
DAEAC-base(ours)	MITDB	88.44%	35.57%	0.51	15.57%	46.93%	0.24
DAEAC-uda(ours)	MITDB	84.37%	56.26%	0.68	23.63%	53.88%	**0.33**

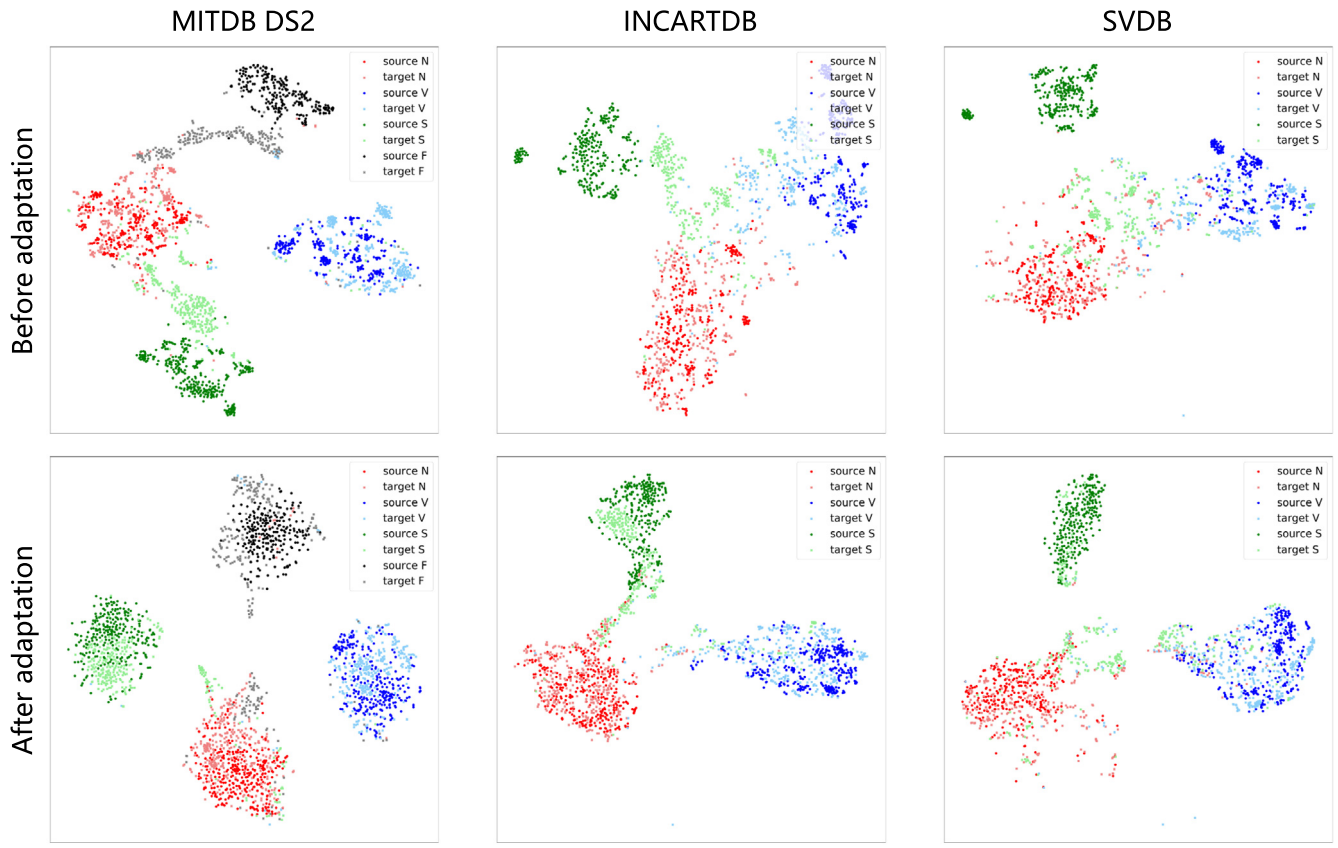


Fig. 6. The t-SNE visualization results of the feature vectors on three databases. The figures in the first row represent the feature distribution before adaptation, and the figures in the second row represent the feature distribution after adaptation. "·" and "×" represnet samples from source domain and target domain, respectively.

tiplying 0.99 every 200 iterations and use the batch size of 256. The training procedure consists of two stages. At the first stage, we train the model with only cross-entropy loss function and source data for 300 epochs. At the second stage, we monitor the training process until the loss functions reach relatively stable values and conduct the training for additional 100 epochs. Totally, 300 epochs are conducted for the adaptation stage. Ultimately, we simply use the last checkpoint for evaluation. Since the input is 1D signal, the size of our model is only 0.09 M. More details can be referred to our codes on https://github.com/chenmingTHU/DAEAC.

5. Experimental results

5.1. Overall performance

In this section, we compare our methods with other state-of-the-art researchs on MITDB, INCARTDB and SVDB and perform visualization experiments to intuitively demonstrate the effectiveness of our method.

5.1.1. Results on MITDB-DS2

The comparison results with our works are shown in Table 2. We mainly compare our method with recently proposed deep learning models [29,5,30,7,8] or related methods [6,14] that shared the same evaluation metrics with us. Both the pre-trained model M2, which serves as the baseline, and adapted model M2 are displayed in the table. Refs. [6,14] used the same idea of unsupervised domain adaptation as us, while they applied patient-specific practice in their work. As we can see in the table, our baseline model DAEAC-base achieves comparable performance on all categories (close or better F1 scores and overall accuracy). After adaptation, DAEAC-uda gains significant performance improvements, and reaches the highest F1 scores on SVEB (0.83) and F (0.65). Compared to the two patient-specific algorithms which trained a specific model for each patient, our method only needs to train a global model without introducing extra storage or computing resources.

5.1.2. Results on INCARTDB and SVDB

The comparison results of cross-dataset experiments are shown in Table 3 and Table 4. As expected, compared to the evaluation results on MITDB DS2, the performances across datasets decrease. After adaptation, the performances have been significantly improved. Specifically, compared to the baseline model, our algorithm improves the F1 score of VEB on INCARTDB and SVDB by 0.06 and 0.12, respectively. Furthermore, it improves SVEB by 0.14 and 0.09, respectively. Our model surpasses other state-of-the-arts in the overall performance of SVEB, and achieves comparable results on VEB.

5.1.3. Visualization

To more intuitively demonstrate the effect of the algorithm on the model, we use t-SNE [32] algorithm to visualize the output features of $\mathcal{F}$. As shown in Fig. 6, after domain adaptation, the distances of the clusters in the two domains are reduced, and the discriminability of each category is also improved. The Cluster-aligning loss aligns the centres of same category from different domains, the separating loss keeps the different clusters away from each other, and the compacting loss makes each cluster more compact. Our algorithm successfully reduces the discrepancy between two different datasets in the feature space, making the learned features in the same category more consistent.

Table 6
Ablation study on SE layers after adaptation.

Method	Overall Accuracy
with SE layers	97.59%
without SE layers	96.90%

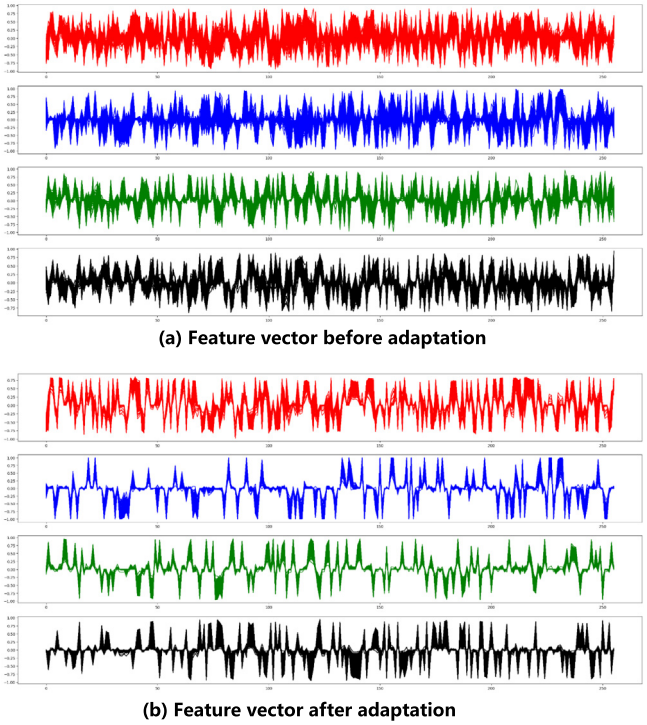


Fig. 7. The visualization of the first four channels in the feature vectors.

Table 5
Ablation study on SE layers and ASPP modules before adaptation.

Database	SE	ASPP	VEB			SVEB			Overall Accuracy
			Se	Pp	F1	Se	Pp	F1	
MITDB	✓	✓	92.01%	89.16%	0.86	71.92%	82.89%	0.77	**94.62%**
		✓	89.24%	88.73%	0.89	58.51%	74.70%	0.66	94.27%
			93.74%	79.38%	0.86	20.93%	56.07%	0.30	89.35%
SVDB	✓	✓	84.02%	37.41%	0.52	14.16%	39.20%	0.21	**84.17%**
		✓	84.74%	29.53%	0.44	16.33%	46.33%	0.24	81.00%
			89.45%	30.07%	0.45	15.27%	49.89%	0.23	80.10%
INCARTDB	✓	✓	77.02%	92.08%	0.84	44.07%	37.38%	0.40	**95.36%**
		✓	76.74%	85.09%	0.81	49.94%	23.69%	0.32	94.03%
			68.56%	87.72%	0.77	63.95%	26.80%	0.38	93.10%

Table 7
Ablation study on items of Cluster-Maintaining loss on MIT-BIH DS2.

Number	$\mathcal{L}_{sep}$	$\mathcal{L}_{comp}$	VEB			SVEB			F			Overall Accuracy
			Se	Pp	F1	Se	Pp	F1	Se	Pp	F1	
1			91.87%	86.11%	0.89	80.56%	84.11%	0.82	27.25%	68.79%	0.39	96.66%
2	✓		94.29%	84.57%	0.89	78.51%	87.89%	0.83	59.55%	61.44%	0.60	96.83%
3		✓	90.75%	92.50%	0.92	75.59%	87.18%	0.81	0.8%	9.6%	0.01	96.85%
4	✓	✓	94.02%	92.30%	0.93	76.52%	90.19%	0.83	57.87%	74.10%	0.65	**97.59%**

Table 8
Impact of the number of unlabeled samples.

Number	VEB			SVEB		
	Se	Pp	F1	Se	Pp	F1
50	96.1%	89.6%	0.93	78.5%	81.1%	0.79
100	96.4%	89.3%	0.93	78.8%	78.2%	0.78
200	95.8%	90.35	0.93	79.4%	90.4%	**0.84**
300	96.6%	90.9%	0.94	78.4%	91.4%	**0.84**
500	96.0%	90.7%	0.93	77.2%	92.9%	**0.84**
1000	96.4%	90.5%	0.93	80.4%	90.5%	**0.85**

5.2. Discussion

In this section, we discuss the effectiveness of the proposed improvement proposals. For the sake of simplicity, we mainly perform ablation studies on $DS1 \rightarrow DS2$ for the multiple regularizers proposed in domain adaptation stage.

5.2.1. The Effectiveness of ASPP Modules and SE Layers

Atrous spatial pooling pyramid (ASPP) module is a multi-field feature extraction block, which uses multiple atrous convolutional layers [33] with different dilation rates to extract feature maps of different receptive fields. To verify the impact of ASPP modules, we replace the modules in Fig. 3 with normal convolutional layers and keep the number of parameters identical. The ablation results are shown in Table 5. The results show that ASPP modules improve the performance on multiple datasets without changing the number of parameters.

Squeeze-and-Excitation layer is an important improvement in our network structure. It has a natural feature selection mechanism that can reweigh different features during training. We ablate the SE layers in Fig. 3 and train the model from scratch on three databases. The comparison results are shown in Table 5. As shown in the Table 5, SE layers improve the performance before adaptation on all three databases. Further, we discuss the impact of SE layers in adaptation stage in Table 6. For simplicity, we verify the impact of SE layers in the adaptation process on DS2. The comparison results prove that SE layers can improve the effectiveness of the adaptation stage, which also indirectly proves that the feature selection mechanism of SE layers is naturally suitable for domain adaptation algorithm. To have an intuitive observation of the impact of SE layers, we visualize several channels of the feature vectors before and after adaptation in Fig. 7. As shown in the figure, SE layers suppress the magnitudes of the redundant elements and amplify the essential ones, illustrating the impact of the channel attention mechanism on the adaptation process.

5.2.2. The effectiveness of regularizers

In this work, we propose three regularizers in the domain adaptation algorithm. The Cluster-aligning loss $\mathcal{L}_{align}$ is the crucial component of the algorithm, while the Cluster-separating loss $\mathcal{L}_{sep}$ and Cluster-compacting loss $\mathcal{L}_{comp}$ are auxiliary components that help the feature learning procedure. We discuss the effectiveness of $\mathcal{L}_{sep}$ and $\mathcal{L}_{comp}$ by removing them from the adaptation stage in turn. The comparison results of four setups are demonstrated in Table 7. Interestingly, $\mathcal{L}_{sep}$ appears to have a significant effect on F, and $\mathcal{L}_{comp}$ shows contribution only when used in conjunction with $\mathcal{L}_{sep}$. This phenomenon might be attributed to the inbalance of different categories. In summary, the proposed regularizers have positive impacts on the adaptation stage according to the experimental results.

5.2.3. The numbers of unlabeled samples

There are two steps in training that need to use pseudo labels. At the beginning of the adaptation stage, cluster centers of all categories in the target domain require initialization. Here, we use a pre-trained model to select samples with confidence scores higher than the pre-defined thresholds. In this step, 17260, 1448, 1424 and 17 pseudo labels for N, V, S, F are selected to initialize the adaptation stage. During the adaptation process, target samples with high confidence scores are selected to update cluster centers. In this step, the number of pseudo labels changes dynamically according to the training process and mini-batch. Roughly, 25% of N samples, and 50% of V and S samples in a mini-batch are selected, and only few F samples are selected due to its tiny amount.

The number of unlabeled samples have a degree of impacts on the final performance. To illustrate the impacts, we conduct a group of brief experiments where we solely change the number of unlabeled beats in each records and conduct the adaptation process for 40 epochs. The results are demonstrated in Table 8. The performance gains (especially for S) increase signally when selecting about 200–300 unlabeled beats, while more unlabeled samples can only bring slight improvement. We can conclude that, when too few unlabeled samples are selected, the prior information provided by abnormal samples is not sufficient for effective adaptation; when too many unlabeled samples are selected, more false positives occur, resulting in slight decreases in precision.

6. Conclusion

In this paper, we propose a novel Domain-Adaptive ECG Arrhythmia Classification model to tackle the inter-patient ECG

heartbeat arrhythmia classification problem. We design a convolutional neural network based on channel attention and multi-scale features to achieve comparable results with other state-of-the-arts with more balanced performance on multiple categories. To mitigate the distribution discrepancies of different records that cause performance degradation, we propose two novel objective functions to enhance deep learning model efficiently. Our method requires no annotations in new records and does not introduce extra computing or storage resources in inference phase. The proposed method can significantly improve the inter-patient performance of deep learning models and can be easily adapted to new data.

There are still some limitations in DAEAC. E.g., when insufficient unlabeled samples are collected, the performance of minor categories may not be effectively enhanced due to the long-tail effect. We will study the problem in our future work.

CRediT authorship contribution statement

Guijin Wang: Conceptualization, Methodology, Resources, Supervision, Funding acquisition, Writing - review & editing. **Ming Chen:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing - review & editing. **Zijian Ding:** Investigation, Writing - original draft, Supervision. **Jiawei Li:** Data curation, Validation, Software. **Ping Zhang:** Project administration, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work is partially funded by the National Key Research and Development Program of China (2017YFB1401804). We also appreciate the support from Beijing National Research Center for Information Science and Technology and Beijing Innovation Center for Future Chip.

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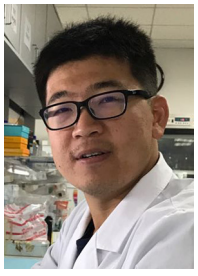


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